

Data Centre Structured Cabling

Rack Unit Standards

Standard 19-inch equipment racks provide 42 to 48 rack units (RU) of vertical space. Patch panels mount at 1 RU per 24 ports. The recommended mounting height for patch panels is between 14 RU and 32 RU to minimise patch cord lengths.

- Standard rack: 19 inch width
- Patch panel density: 24 ports per 1 RU
- Optimal height: 14 RU to 32 RU

Top-of-Rack Architecture

Top-of-rack (ToR) switching places the network switch at the top of each server cabinet. Server-to-switch cabling is limited to 5 metres. This architecture reduces horizontal cable management congestion compared to end-of-row switching.

- Server-to-switch: 5 m maximum
- Benefit: reduced cable management congestion

Hot Aisle / Cold Aisle

Data centre layouts shall orient cabinets in alternating hot-aisle and cold-aisle configurations. Cold aisles supply conditioned air to equipment intakes. Hot aisles collect exhaust air for return to the computer room air handler (CRAH).